

Data Detective

Every graded set, every week — 3 weeks, 15 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Reading a Graph

Each question describes a graph in words. Find the scale before you answer anything.

Warm-Up 12 ITEMS • RECALL

1 Each ★ = 5 books.
Monday has 3 ★ — books

2 Each ★ = 5 books. Friday
has 6 ★ — books

3 A bar reaches 40 on a
scale marked every 10 — its
value

4 Each ■ = 2 goals. A bar of
7 ■ — goals

5 Bars: Red 8, Blue 5 —
total

6 Bars: Red 8, Blue 5 —
difference

7 Each ★ = 5 books. 4 ★ —
books

8 Each ★ = 5. 9 ★

9 Each ■ = 2 goals. 7 ■

10 A bar chart marked
every 5, a bar at the second
mark

11 Each ★ = 10. 6 ★

12 Each ■ = 4. 5 ■

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Scale every 5. A bar halfway between 20 and 30 (Halfway is one mark up)

.....

14. Three full groups of five tallies plus 2 — total (15 and 2 more)

.....

15. Graph shows Mon 12, Tue 9, Wed 15 — three-day total

.....

16. Each ■ = 2 goals. 24 goals is how many ■ (24 ÷ 2)

.....

17. Each ★ = 4 pets. 6 ★ and a half ★ — pets (Half a star is 2)

.....

18. Each ★ = 5. Half a ★

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Mon 12, Tue 9, Wed 15, Thu 0 — mean of the four days ($36 \div 4$ — the zero still counts as a day)

24. Each ★ = 5. Four rows total 90 books — stars altogether ($90 \div 5$)

25. 90 items with a symbol worth 12 — how many symbols, to one decimal

26. A bar reaching 3.5 marks on a scale of 20

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mean, Median, Mode, Range

Every set here is chosen so the answer comes out whole. Put the numbers in order before you look for a middle.

Warm-Up 4 ITEMS • RECALL

1 Range of 3, 9, 12

2 Mode of 5, 7, 7, 9

3 Median of 2, 5, 9

4 Total of 6, 7, 8, 9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

5. Median of 4, 7, 7, 9, 13

.....

6. Range of 4, 7, 7, 9, 13 (13 - 4)

.....

7. Median of 2, 4, 6, 8 (Halfway between the two middles)

.....

When the Average Lies

One extreme value. Watch carefully which measures move and which do not.

Warm-Up 10 ITEMS • RECALL

1 Median of 5, 5, 5, 5

2 Median of 5, 5, 5, 25

3 Range of 5, 5, 5, 25

4 Mode of 5, 5, 5, 25

5 Median of 1, 3, 5

6 Median of 4, 8, 12

7 Median of 2, 2, 9

8 Mode of 3, 3, 7

9 Mode of 6, 6, 6, 1

10 Range of 2, 9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Those same five people — median

12. Those same five people — range

13. Type the value of the measure that best describes a typical one of those five people
(The median, 10 — the mean of 20 describes nobody)

14. Median of 3, 8, 1, 9, 5

15. Median of 2, 4, 6, 8

16. Mode of 4, 7, 7, 9, 9, 9



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Chance as a Number

Chance is a count out of a total. Type the count that is asked for.

Warm-Up 12 ITEMS • RECALL

1 Faces on one die

2 Ways to roll a 4 on one die

3 Even numbers on one die

4 Ways to roll more than 4 on one die

5 Bag of 3 red and 5 blue — total counters

6 That bag — how many draws out of 8 should be red

7 Faces on a die

8 Ways to roll a 2

9 Odd numbers on a die

10 Ways to roll less than 3

11 A bag of 2 red and 6 blue — total

12 Sides on a coin

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Chance of an even roll is _ out of 6 — type the top number

.....

14. 60 rolls of a die — expected number of 3s ($60 \div 6$)

.....

15. Bag of 3 red, 5 blue. 40 draws, replacing each time — expected reds (3 in every 8, and 40 is five eights)

.....

16. A coin flipped 50 times — expected heads

.....

17. Spinner in 8 equal parts, 2 gold. 40 spins — expected golds (2 in every 8)

.....

18. 36 rolls of a die — expected 2s

.....

Predict the Roll & Ask 20 People

Enrichment day. Predict first, in writing, every single time.

✦ Warm-Up 4 ITEMS • RECALL

1 People in the survey

2 Trials in Predict the Roll

3 Median of 1, 5, 9

4 20 people, 15 say yes — percent saying yes

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 20 people, 4 pick blue. Out of 100 people, expect (15 out of 20 is 75 out of 100)

6. Range of 5, 22, 13 (Multiply both by 5)

7. 50 rolls of a die — expected number of 6s

8. 20 people own a mean of 6 pets each — total pets

Challenge NOT GRADED • REASON IT OUT

9. You predicted 8 sixes and rolled 14. Difference(50 ÷ 6 is 8 with 2 over)

10. Mean of 4, 6

11. Mean of 7, 9, 11, 13 (Real data misses the prediction. That is not an error.)

12. 5, 5, 5, 5, 30 — the mean

13. That same set — the median

14. Which describes it better — type mean or median

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Reading a Graph

OUT OF 18

Warm-Up • 1-12 • 15 | 30 | 40 | 14 | 13 | 3 | 20 | 45 | 14 | 10 | 60 | 20

Core • 13-18 • 25 | 17 | 36 | 12 | 26 | 2.5

Challenge • 23-26 • 9 | 18 | 7.5 | 70

1.2 • Mean, Median, Mode, Range

OUT OF 7

Warm-Up • 1-4 • 9 | 7 | 5 | 30

Core • 5-7 • 7 | 9 | 5

Challenge • 8-19 • 6 | 15 | 8 | 15 | 50 | 8 | 4 | 15 | 5 | 2.5 | 10 | 5

1.3 • When the Average Lies

OUT OF 16

Warm-Up • 1-10 • 5 | 5 | 20 | 5 | 3 | 8 | 2 | 3 | 6 | 7

Core • 11-16 • 10 | 50 | 10 | 5 | 5 | 9

Challenge • 19-26 • 5 | 10 | 20 | 10 | 300 | 200 | 50 | 2

1.4 • Chance as a Number

OUT OF 18

Warm-Up • 1-12 • 6 | 1 | 3 | 2 | 8 | 3 | 6 | 1 | 3 | 2 | 8 | 2

Core • 13-18 • 3 | 10 | 15 | 25 | 10 | 6

Challenge • 23-26 • 6 | 36 | 5 | 1

Fri • Predict the Roll & Ask 20 People

OUT OF 8

Warm-Up • 1-4 • 20 | 50 | 5 | 5

Core • 5-8 • 75 | 20 | 10 | 17

Challenge • 9-14 • 8 | 120 | 6 | 10 | 5 | median

Write the Question

A fair question doesn't push the answer. Rewrite the loaded ones.

Warm-Up 10 ITEMS • RECALL

1 People you will ask

2 Answer choices in a yes/no question

3 20 people, 10 say yes — percent saying yes

4 20 people, 5 say yes — percent

5 20 people, 20 say yes — percent

6 20 people, 0 say yes — percent

7 20 people, 4 say yes — percent

8 20 people, 16 say yes — percent

9 20 people, 2 say yes — percent

10 20 people, 18 say yes — percent

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 20 people, 15 say yes — percent

.....

12. 20 people, 12 say yes — percent

.....

13. 20 people, 3 say yes — percent

.....

14. If 60% of 20 said yes, how many people

.....

15. 20 people across 4 choices, evenly — each choice

.....

16. 20 people, 13 say yes — percent

.....

Ask 20 People

Tally as they answer. No tidying up afterwards.

Warm-Up 12 ITEMS • RECALL

1 Four tally marks plus a cross-stroke — the count

2 Two groups of five

3 Three groups of five plus 2

4 Four groups of five

5 One group of five plus 3

6 20 minus 13

7 Five tally marks — the count

8 Three groups of five

9 Two groups of five plus 4

10 Four groups of five plus 1

11 One group of five plus 1

12 20 minus 11

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Counts 8, 6, 4, 2 — the total

.....

14. Those counts — the largest

.....

15. Those counts — the range

.....

16. Counts 9, 5, 6 — the total

.....

17. If three choices total 20 and two are 9 and 5 — the third

.....

18. Counts 7, 6, 4, 3 — the total

.....

★ Challenge NOT GRADED • REASON IT OUT

21. 20 answers over 4 choices, one has 11 — the other three total

22. Counts 8, 6, 4, 2 — the mean

23. 20 answers over 5 choices, one has 8 — the other four total

24. Counts 7, 6, 4, 3 — the mean

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Graph What You Got

Bar graph of your own data, plus all four measures.

Warm-Up 9 ITEMS • RECALL

1 Counts 8, 6, 4, 2 — the total

2 Their range

3 The mode of 5, 7, 7, 9

4 The median of 2, 5, 9

5 A scale marked every 2 — the value 3 marks up

6 Counts 10, 5, 3, 2 — the total

7 The mode of 4, 9, 9, 12

8 The median of 3, 7, 11

9 A scale marked every 5 — the value 4 marks up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Those counts — the median

.....

11. Those counts — the range

.....

12. Water 8, juice 6, milk 4, tea 2 — the mode

.....

13. Out of 100 people, expect how many water

.....

14. Pizza 10, pasta 5, rice 3, soup 2 — the mode

.....

15. Out of 100 people, expect how many pizza

.....

Challenge NOT GRADED • REASON IT OUT

16. Their mean

17. Counts 12, 8, 6, 4 — the mean

18. Counts 9, 5, 3, 2, 1 — the mean

19. Counts 14, 10, 8, 4 — the mean

20. Counts 8, 6, 3, 2, 1 — the mean

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Chance as a Number

Chance is a count out of a total. Nothing more mysterious.

Warm-Up 12 ITEMS • RECALL

1 Faces on one die

2 Ways to roll a 4

3 Even numbers on a die

4 Ways to roll more than 4

5 A bag of 3 red and 5 blue
— total

6 Sides on a coin

7 Faces on two dice added
— the smallest total

8 The largest total on two
dice

9 Ways to roll a 6 on one
die

10 Numbers under 3 on a
die

11 A bag of 5 red and 5
blue — total

12 Ways to flip tails

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 60 rolls of a die — expected 3s

.....

14. A coin flipped 50 times — expected heads

.....

15. 3 red in 8, drawn 40 times — expected reds

.....

16. A spinner of 8 parts, 2 gold, 40 spins — expected golds

.....

17. A bag of 4 red and 6 blue — reds expected in 20 draws

.....

18. 120 rolls of a die — expected 4s

.....

Mid-Unit Quiz

Eight items across Weeks 1–2. 85% to keep flying.

Warm-Up 4 ITEMS • RECALL

1 Each ★ = 5 books. 7 ★ — books

2 Each ★ = 10 books. 4 ★ — books

3 Median of 11, 4, 7, 20, 9

4 Mode and range of 3, 8, 8, 8, 15 — type the range

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. A bag of 4 red and 6 blue — reds in 10 draws

6. 60 rolls — expected 5s

7. Median of 3, 9, 15, 21, 27

8. Range of 2, 14, 8, 20

9. A bag of 3 red and 7 blue — reds in 20 draws

10. Mean of 6, 8, 10, 12

11. Six numbers with mean 9 — their total

12. 2, 2, 2, 2, 22 — the mean

13. Mean of 5, 10, 15

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Write the Question

OUT OF 16

Warm-Up • 1-10 • 20 | 2 | 50 | 25 | 100 | 0 | 20 | 80 | 10 | 90

Core • 11-16 • 75 | 60 | 15 | 12 | 5 | 65

Challenge • 21-24 • 35 | 35 | 85 | 170

2.2 • Ask 20 People

OUT OF 18

Warm-Up • 1-12 • 5 | 10 | 17 | 20 | 8 | 7 | 5 | 15 | 14 | 21 | 6 | 9

Core • 13-18 • 20 | 8 | 6 | 20 | 6 | 20

Challenge • 21-24 • 9 | 5 | 12 | 5

2.3 • Graph What You Got

OUT OF 15

Warm-Up • 1-9 • 20 | 6 | 7 | 5 | 6 | 20 | 9 | 7 | 20

Core • 10-15 • 7 | 8 | water | 40 | pizza | 50

Challenge • 16-20 • 5 | 7.5 | 4 | 9 | 4

2.4 • Chance as a Number

OUT OF 18

Warm-Up • 1-12 • 6 | 1 | 3 | 2 | 8 | 2 | 2 | 12 | 1 | 2 | 10 | 1

Core • 13-18 • 10 | 25 | 15 | 10 | 8 | 20

Challenge • 23-26 • 6 | 36 | 5 | 2

Fri • Mid-Unit Quiz

OUT OF 13

Warm-Up • 1-4 • 35 | 9 | 40 | 10

Core • 5-13 • 9 | 12 | 54 | 4 | 10 | 15 | 18 | 56 | 6

Challenge • 14-16 • 6 | 20 | 9

Publish the Findings

One headline, one graph, one honest limitation.

Warm-Up 11 ITEMS • RECALL

1 People surveyed

2 15 of 20 — the percent

3 5 of 20 — the percent

4 Out of 100, expect how many if 15 of 20 said yes

5 8 of 20 — the percent

6 20 minus 8

7 14 of 20 — the percent

8 6 of 20 — the percent

9 Out of 100, expect how many if 14 of 20 said yes

10 7 of 20 — the percent

11 20 minus 14

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. 12 of 20 — out of 100

.....

13. A headline says most people. The smallest count of 20 that earns it

.....

14. 9 of 20 — the percent

.....

15. Does 45% earn most — yes or no

.....

16. 17 of 20 — the percent

.....

17. 11 of 20 — out of 100

.....

The Misleading Graph

Start the axis at 90 and watch a tiny gap look enormous.

Warm-Up 12 ITEMS • RECALL

1 Bars at 95 and 98 — the true difference

2 Bars at 50 and 60 — the difference

3 An axis starting at 0 shows a bar of 95 as how tall, in units

4 An axis starting at 90 shows that bar as

5 Bars at 20 and 40 — the difference

6 40 is how many times 20

7 Bars at 90 and 94 — the true difference

8 Bars at 40 and 60 — the difference

9 An axis from 0 shows a bar of 90 as how tall, in units

10 An axis from 80 shows that bar as

11 Bars at 10 and 30 — the difference

12 30 is how many times 10

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Bars at 95 and 98 on an axis from 90 — the drawn heights differ by

.....

14. Drawn from 90, 98 appears how many times taller than 95

.....

15. An axis starting at 0 is honest — yes or no

.....

16. Bars at 102 and 104 from an axis at 100 — apparent times taller

.....

17. Bars at 90 and 94 on an axis from 80 — drawn heights differ by

.....

18. Drawn from 80, 94 appears how many times taller than 90

.....

★ Challenge NOT GRADED • REASON IT OUT

20. The true ratio of 98 to 95, to one decimal

21. Sales of 1,020 and 1,040 drawn from 1,000 — apparent times taller

22. Their true percent difference, to the nearest whole

23. The true ratio of 94 to 90, to one decimal

24. Sales of 2,020 and 2,060 drawn from 2,000 — apparent times taller

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Fifty Trials

Predict, roll, tally, and explain the gap in writing.

Warm-Up 10 ITEMS • RECALL

1 Trials

2 Faces on a die

3 $50 \div 6$ — the whole part

4 A coin, 50 flips —
expected heads

5 Six faces, expected each
in 60 rolls

6 Sides on a coin

7 $50 \div 2$

8 Six faces, expected each
in 120 rolls

9 A coin, 100 flips —
expected heads

10 $50 \div 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Predicted 8, rolled 14 — the difference

.....

12. Predicted 8, rolled 3 — the difference

.....

13. 600 rolls — expected 6s

.....

14. Is a spread of a few either way normal in 50 rolls — yes or no

.....

15. Six tallies totalling 50, five are 8 — the sixth

.....

16. Predicted 8, rolled 12 — the difference

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 Median of 2, 5, 9

2 Mode of 5, 7, 7, 9

3 Range of 3, 9, 12

4 Each ★ = 5. 6 ★ — books

5 Even numbers on a die

6 Median of 4, 7, 10

7 Mode of 2, 8, 8, 11

8 Range of 5, 11, 17

9 Each ★ = 10. 7 ★ — books

10 Odd numbers on a die

11 Median of that set

12 60 rolls — expected 3s

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 15 of 20 — the percent

.....

14. 120 rolls — expected 5s

.....

15. 14 of 20 — the percent

.....

16. Mean of 4, 6, 8

.....

17. Mean of 4, 7, 7, 9, 13

.....

18. Five numbers with mean 10 — their total

.....

19. Six houses at 200 and one at 900 — the mean

.....

20. Those same houses — the median

.....

21. Mean of 6, 10, 14

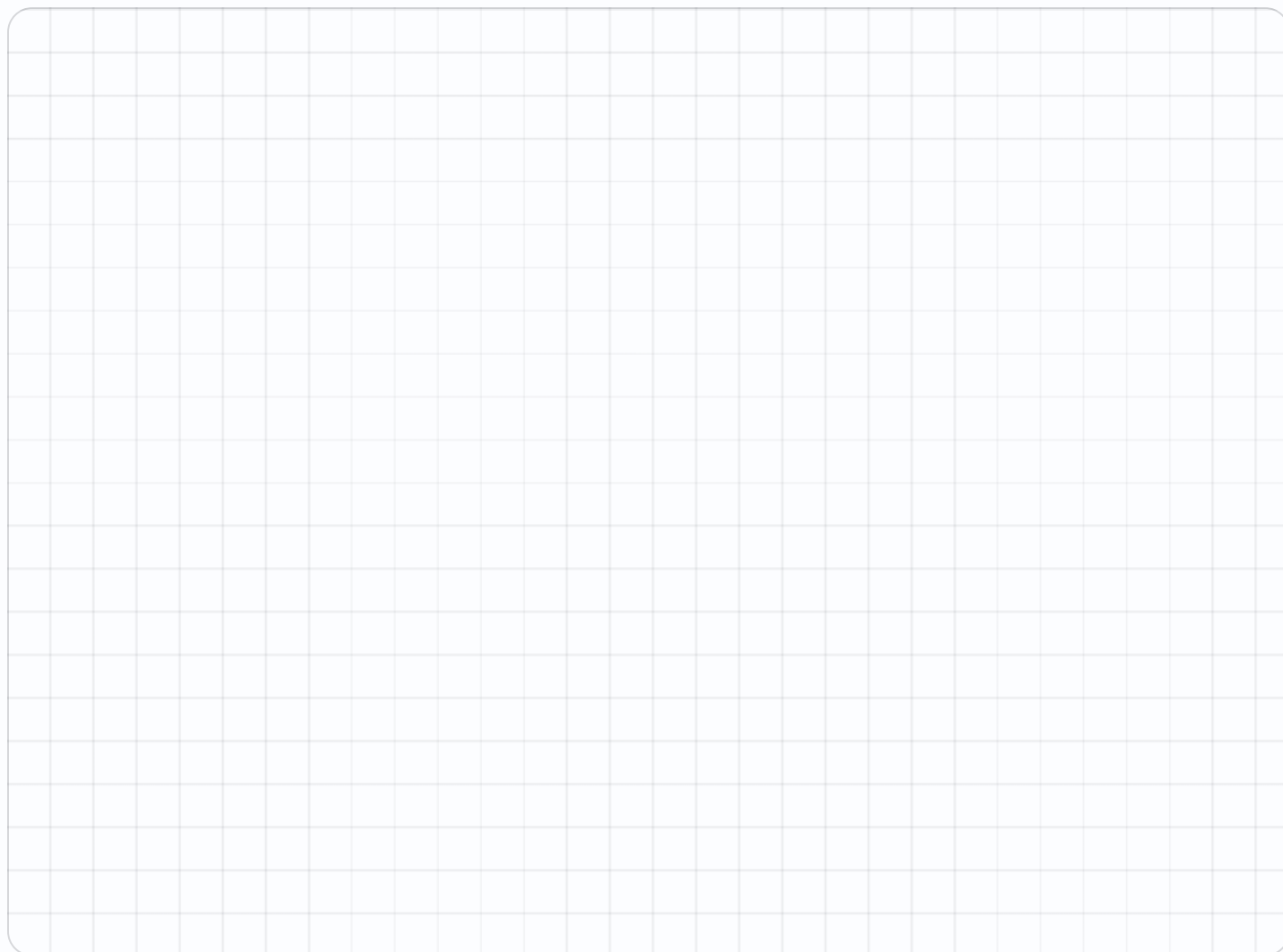
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★ Challenge NOT GRADED • REASON IT OUT

22. Mean of 6, 8, 8, 10, 13

23. Six numbers with mean 12 — their total

24. Nine houses at 150 and one at 1,050 — the mean

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mission 07 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 A bar chart marked every 4, a bar at the third mark — its value

2 Each ■ = 2 goals. 9 ■ — goals

3 A bar chart marked every 5, a bar at the fourth mark

4 Each ■ = 3 goals. 8 ■

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Median of 5, 12, 3, 8, 21, 9

.....

6. Range of 7, 7, 19, 4, 11

.....

7. A spinner of 8 parts, 2 gold, 40 spins — expected golds

.....

8. 20 people, 8 pick water — out of 100

.....

9. Those four — the mode

.....

10. Median of 4, 11, 6, 15, 9, 20

.....

11. Range of 6, 6, 22, 3, 14

.....

12. A spinner of 5 parts, 2 gold, 50 spins — expected golds

.....

13. 20 people, 7 pick tea — out of 100

.....

14. Mean of 14, 16, 18, 20

.....

15. Five scores total 45, a sixth of 9 added — new mean

.....

16. Water 8, juice 6, milk 4, tea 2 — the mean

.....

17. Two dice — ways to total 7

.....

18. 2, 2, 2, 2, 22 — mean and median; type the median

.....

19. Mean of 10, 20, 30, 40

.....



Working space



Error journal

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Publish the Findings

OUT OF 17

Warm-Up • 1-11 • 20 | 75 | 25 | 75 | 40 | 12 | 70 | 30 | 70 | 35 | 6

Core • 12-17 • 60 | 11 | 45 | no | 85 | 55

Challenge • 21-24 • 250 | 3750 | 400 | 5600

3.2 • The Misleading Graph

OUT OF 18

Warm-Up • 1-12 • 3 | 10 | 95 | 5 | 20 | 2 | 4 | 20 | 90 | 10 | 20 | 3

Core • 13-18 • 3 | 1.6 | yes | 2 | 4 | 1.4

Challenge • 20-24 • 1 | 2 | 2 | 1 | 3

3.3 • Fifty Trials

OUT OF 16

Warm-Up • 1-10 • 50 | 6 | 8 | 25 | 10 | 2 | 25 | 20 | 50 | 10

Core • 11-16 • 6 | 5 | 100 | yes | 10 | 4

Challenge • 20-24 • 2 | 6 | 1 | 12 | 2

Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-12 • 6 | 5 | 7 | 9 | 30 | 3 | 10 | 7 | 8 | 12 | 70 | 3

Core • 13-21 • 8 | 7 | 50 | 10 | 75 | 9 | 72 | 20 | 70

Challenge • 22-24 • 300 | 200 | 240

Fri • Mission 07 Test

OUT OF 19

Warm-Up • 1-4 • 12 | 18 | 20 | 24

Core • 5-19 • 17 | 8.5 | 15 | 9 | 10 | 40 | 5 | water | 25 | 10 | 19 | 9 | 20
| 35 | 5

Challenge • 20-23 • 6 | 2 | 4 | 3